CS411W Lab 1

Take The Lead Product Description

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Figure 1: Company Logo

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1. Introduction

Veterinary care is one of the few necessary expenses that most families never plan for. As of March 2026, roughly 94 million households in the United States own at least one pet, yet only about 6.4 million pets are insured (Insurance Information Institute, 2026). Insurance enrollment has grown in recent years, but the gap means that most owners pay for emergencies and routine care entirely out of pocket. When a bill arrives that a family cannot cover, treatment is delayed, declined, or replaced by surrender or euthanasia, and the pet is the one who pays for the shortfall.

The cost problem is misunderstood on both sides of the examination table. Owners rarely know why services are priced the way they are, which options exist to pay for them, or which questions to ask before agreeing to treatment. Veterinary staff absorb the emotional weight of those conversations. Veterinarians are 1.6 times (men) and 2.4 times (women) more likely to die by suicide than the general population, a pattern associated with compassion fatigue from repeated exposure to suffering and difficult financial decisions (Nett et al., 2019). Existing assistance programs such as The Pet Fund (2026), Frankie's Friends (2026), and Treasured Memories Community Funding (2019) help, but each is limited in funding, scope, or speed, and none combines fundraising, education, and payment coordination in one place.

Take The Lead (TTL) is a web application that addresses this problem directly. A pet owner, called a member, creates a 30-day donation campaign after a 48-hour preparation window. Working with the participating emergency or general veterinary hospital, Take the Lead either arranges a delay in payment or supplies a Minimum Down Payment so the pet can be treated immediately rather than after the money is raised. During the campaign the Member shares a single link so friends, family, and the public can donate. When the campaign ends, proceeds go toward the bill; any excess is returned to the Member, and any remaining balance is

the Member's responsibility under the agreement. Because that balance is due at the end of the campaign, Take the Lead also provides information on pet insurance, pet-specific credit, assistance organizations, and the realities of veterinary pricing *before* a member commits, along with an AI chatbot that directs users to helpful links, phone numbers, and application-provided resources.

In short, Take the Lead is a response to the problem of affording emergencies and standard animal care that also works to keep owners and veterinary staff informed and on the same side. The following sections describe the product, its key features and major components, and the case study that will guide the prototype.

2. Take the Lead Product Description

Take The Lead is a website application that helps families obtain veterinary care by providing a Promise-to-Pay agreement or a Minimum Down Payment (MDP) to the veterinary hospital while giving the Member a 30-day window to raise the remaining funds, either through the application's donation service or through other means. The Member runs a Story, the campaign's narrative and media, for 30 days by sharing its link on social media and messaging platforms so that friends, family, and acquaintances can contribute.

When the 30-day campaign ends, the funds generated are distributed in a fixed order. If Take the Lead supplied a Minimum Down Payment, it is repaid first; the veterinary hospital is paid next; and anything left is returned to the Member, who may keep the funds or donate them back to Take the Lead to support future campaigns. If a campaign does not generate sufficient funds, the remaining balance is owed to Take the Lead (if an MDP was supplied) and to the veterinary hospital according to the Promise-to-Pay agreement. The goal is that no pet waits for treatment because of a bill, and that no Member is surprised by what they owe.

The application serves three groups. Members are registered pet owners who create and manage campaigns. Guests are visitors who can read Stories, donate, use the chatbot, and browse resources without an account. Participating in veterinary hospitals verify the anticipated cost of treatment and accept the payment arrangement. To help Members with any remaining balance, Take the Lead also offers reading and learning material: why services are necessary and why their expense is not the fault of the staff, information that builds compassion for veterinary workers, and guidance on pet insurance and pet-only credit cards. The key features that deliver these goals are described next, followed by the major components that make them work.

2.1. Key Product Features and Capabilities

Fundraising campaigns (Stories). A Member creates a campaign that contains the pet's Story, a photo, the participating veterinary hospital, and the bill amount. A 48-hour preparation window lets the Member finish the Story before it goes live; the campaign then runs for 30 days. Each campaign has one shareable link, a progress bar, days remaining, and a list of supporters, so the Member and donors can see momentum immediately.

Promise-to-Pay agreements and Minimum Down Payments. Take The Lead works in tandem with veterinary hospitals so that treatment begins immediately. Depending on the hospital and the case, Take the Lead either pays an agreed Minimum Down Payment up front or enters an agreement with the hospital for the 30-day period while the Member raises funds. The terms, including the most a member could still owe, are presented in plain language and must be accepted before a campaign is published.

Donations and fund distribution. Guests and Members can donate under their name or anonymously and leave a message of support that the Member approves before it appears. Donations are applied directly to the veterinary bill in the order described in Section 2. The prototype processes payments through a test-mode payment service so that no real money moves

during development.

AI chatbot for funding research and care guidance. The chatbot is Take The Lead's second key feature. Given the pet, condition, estimated cost, and ZIP code, it searches the application's database of grants, nonprofits, and financing options and returns the ones an owner may qualify for, each with a link to its source. When an owner describes symptoms, a rules-based safety check runs first: red-flag symptoms such as trouble breathing, collapse, or possible poisoning return an immediate instruction to go to an emergency hospital, with the nearest facilities by ZIP code. The chatbot then explains how urgent the situation appears, where to go, and what to expect. It does not diagnose, does not create campaigns, and is not a substitute for veterinary advice.

Information resources. Rather than be a crutch, Take the Lead supplies the information that helps families avoid the next crisis: how pet insurance works and where to get it, pet-only credit lines, why routine and preventative care matter, and why veterinary services cost what they do. It also shares the compassion-fatigue and suicide-rate data behind the profession, with links to Not One More Vet (2025) and similar organizations, to build understanding between clients and staff.

Veterinary hospital and emergency directory. A directory of participating hospitals, emergency rooms, and urgent care facilities, searchable by ZIP code, supports both the chatbot and users who simply need to know where to go. These features are delivered by the components described in the next section.

2.2. Major Components (Hardware/Software)

Take The Lead is organized into four functional areas that communicate over HTTPS, as shown in the Major Functional Component Diagram in Figure 2: the users and their devices, the frontend, the backend application layer, and external services, with a single data store beneath

the backend.

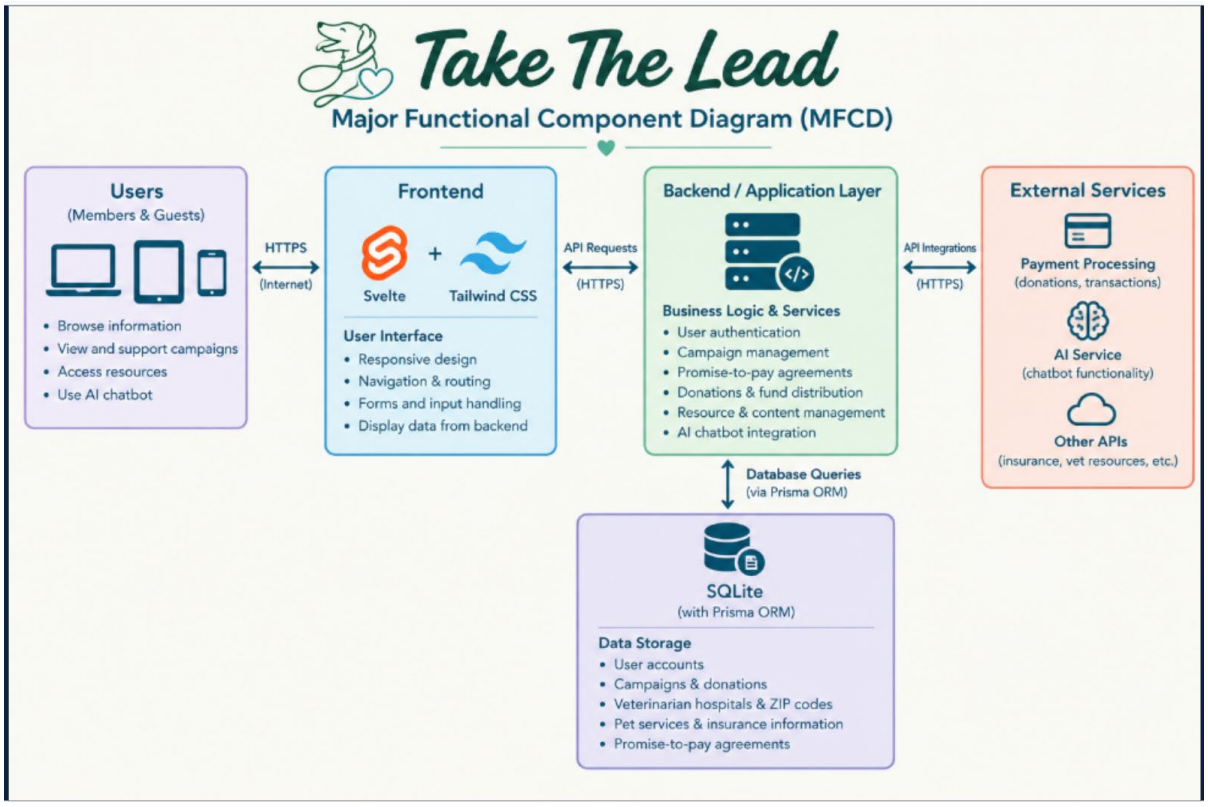


Figure 2: Take the Lead Major Functional Component Diagram.

Hardware. No specialized hardware is required. Members and Guests use any desktop computer, tablet, or smartphone with a modern web browser. The application and its database run on a server hosted by Old Dominion University, which also hosts the AI service used by the chatbot. Participating in veterinary hospitals interact through the same web interface rather than through dedicated equipment.

Frontend. The user interface is built with Svelte and styled with Tailwind CSS. It provides responsive layouts for each device size, navigation and routing between the home page, campaign pages, dashboard, chat, and resources, form and input handling for campaign creation and donations, and display of data returned by the backend.

Backend / application layer. The backend holds the business logic and services: user authentication for Members and Guests, campaign management including the 48-hour and 30-

day windows, Promise-to-Pay agreements, donations and fund distribution, resource and content management, and integration with the AI chatbot. The frontend reaches these services through HTTPS API requests.

Data storage. Persistent data lives in a SQLite database accessed through Prisma ORM. It stores user accounts, campaigns and donations, veterinary hospitals and ZIP codes, pet services and insurance information, and Promise-to-Pay agreements. Seed data provides demonstration accounts, hospitals, campaigns, and funding sources so every teammate and every demonstration starts from the same state.

External services. Three kinds of outside services are reached by API integration over HTTPS: a payment processing service for donations and transactions (Stripe in test mode for the prototype), the AI service that powers the chatbot, and other APIs or data sources for insurance and veterinary resources. With the components in place, the next section identifies the case study that the prototype will serve.

3. Identification of Case Study

Take The Lead is being developed for two groups whose interests are usually treated separately: the people who own pets and the people who treat them. This region provides a realistic mix of emergency, general, and low-cost facilities within a short distance, which the ZIP-code directory and chatbot are designed around.

For pet owners, the product exists to make veterinary care affordable when it is needed most: emergencies, urgent care, medication, and medicated or health-specific food. It provides learning materials that promote a healthier client-veterinarian relationship, a clearer understanding of services and their costs, an explanation of why those costs generally do not lie with the staff, and practical information on pet insurance, including where to get it and how some

employers offer it alongside human health coverage. By encouraging continued wellness care and reducing the number of animals surrendered for lack of funds, the product aims to improve outcomes for pets as well as their families.

For veterinarians and their staff, the product is meant to combat compassion fatigue, one of the leading causes of leaving the profession or of self-harm (Nett et al., 2019). It does so by removing the owner's inability to pay from the examination room, by driving revenue toward wellness and preventative visits rather than only emergencies, and by linking staff and clients to Not One More Vet (2025) and other institutions that offer help.

Beyond the initial case study, the same platform could serve animal shelters and rescues, veterinary teaching hospitals, and pet owners in other regions by adding facilities to the directory and funding sources to the database. The prototype will demonstrate the campaign workflow, the Promise-to-Pay terms, simulated and test-mode payments, and the chatbot's funding and care guidance using seeded data, which is the foundation for the real-world product described above.

4. Glossary

AI Chatbot – The assistant inside Take the Lead that answers questions, finds funding sources, and offers care guidance by pointing users to links, phone numbers, and application-provided information.

Campaign – See Story. The 30-day fundraising period a member runs for a specific veterinary bill, including the 48-hour preparation window before it.

Compassion Fatigue – The emotional and physical exhaustion that builds up in veterinary staff from constant exposure to sick animals, grieving owners, and difficult cost decisions.

ER (Emergency Room) – A veterinary hospital equipped for life-threatening cases, usually open around the clock and priced accordingly.

Guest – A visitor without an account who can read Stories, donate, use the chatbot, and browse resources.

MDP (Minimum Down Payment) – The up-front amount, capped at $5,000, that Take the Lead supplies to a veterinary hospital so treatment can begin before the campaign has raised money.

Member – A user with a registered account on the web application "Take the Lead" who can create and manage a campaign.

MFCD (Major Functional Component Diagram) – The diagram (Figure 1) that shows the application's main parts and how data moves between them.

NOMV (Not One More Vet) – A nonprofit that supports the mental health of veterinary professionals.

Preventative Care – Animal care meant to prevent illness, such as vaccines, check-ups, and joint x-rays for older animals; also called wellness care.

Promise-to-Pay Agreement – The arrangement among the Member, the veterinary hospital, and Take the Lead that states who pays what, when, and the most the Member could still owe after

the campaign.

Story – The 30-day campaign that a member runs, together with the written narrative, photos, and bill details that explain the need; interchangeable with "campaign."

TTL – Short-hand for the application and organization Take the Lead.

UC (Urgent Care) – A veterinary facility for problems that need same-day attention but are not immediately life-threatening; typically, fewer hours and services than an ER.

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